

Question number	Answer	Mark
6(a)(i)	The only correct answer is D <i>A is incorrect because diffusion does not use proteins</i> <i>B is incorrect because diffusion does not use proteins</i> <i>C is incorrect because diffusion does not use proteins</i>	(1)

Question number	Answer	Mark
6(a)(ii)	The only correct answer is B <i>A is incorrect because diffusion does not use energy</i> <i>C is incorrect because facilitated diffusion does not use energy</i> <i>D is incorrect because facilitated diffusion does not use energy</i>	(1)

Question number	Answer	Mark
6(a)(iii)	The only correct answer is B <i>A is incorrect because solutes can move against their concentration gradient in active transport</i> <i>C is incorrect because solutes can move against their concentration gradient in active transport</i> <i>D is incorrect because solutes can move against their concentration gradient in active transport</i>	(1)

Question number	Answer	Additional guidance	Mark
6(b)	An explanation that includes the following points: • (free water molecules) because some water molecules are {restricted / prevented} from movement (1) • (partially permeable membrane} because membrane allows some (types of) molecules to pass through it (1) • (down a water potential gradient) because water molecules move {from a dilute solution to a more concentrated one / from a high osmotic potential to a lower one / from a high solute potential to a lower one} (1)	ACCEPT a description ACCEPT from a low concentration of solute to a high concentration of solute from hypotonic to hypertonic solution IGNORE from a high water concentration to a lower one	(3)

Question number	Answer	Additional guidance	Mark
*6(c)	Indicative content: • water content decreases when crabs first moved (D) • because water passes out of cells by {osmosis / description of osmosis} (E) • water content then increases gradually (D) • because of the increase in amino acid concentration (E) • amino acid increases (D) • sharply at first and then rate decreases (D) • because {insoluble / muscle} proteins broken down into amino acids (E) • by hydrolysis (E) • because crab eats more {protein / amino acids} (E) • which is digested into amino acids (E) • which are taken up into the cells by {active transport / facilitated diffusion} (from the blood) (E) • water content increases because the amino acid content increase (E) • which makes the cytoplasm more concentrated than the sea water (E) • therefore crabs do not die from dehydration (E)	Level 1 1 mark = simple description of one graph 2 marks = simple description of both graphs Level 2 3 marks = and a simple explanation of one graph 4 marks = and a simple explanation of both graphs Level 3 5 marks = a simple explanation of one graph and a more detailed explanation for the other graph 6 marks = a more detailed explanation for both graphs	(6)